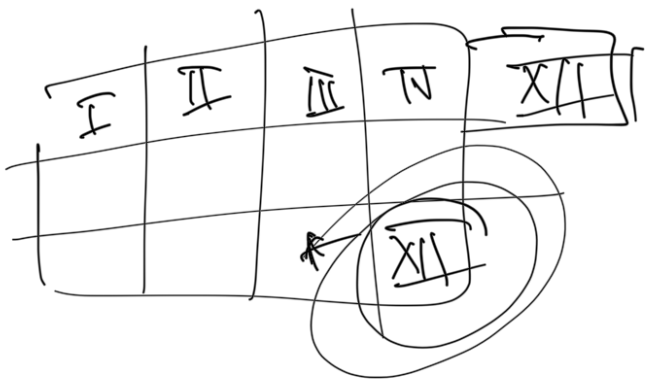
Value => The information that one transformer sub-block will passing on to another transformer sub-block.





FFN

$$y^{out} = \text{gelu}(W_1 x^{in} + b_1) + b_2$$

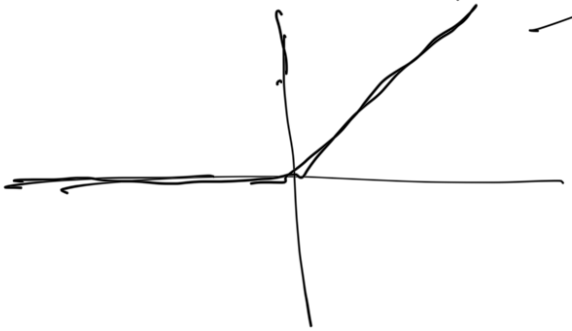
Gelu (Gaussian Error Linear Unit)



Relu

$$y = x\,\phi(x)$$

$$\phi(x) = 0 \quad x < 0$$
$$= 1 \quad x > 0$$



$$\tanh \Rightarrow \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$



Gelu (Gaussian Error linear unit)

$$y = x\,\phi(x)$$

$$\phi(x) = P(X \le x)$$

X is a rv

$$\Downarrow$$

$$X \sim N(0,1)$$

$$\phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,dt$$

of Parameters $\Rightarrow$ very large in Transformers

for effective training

Chain of operations should be differentiable

Gelu

$y = x\,\phi(x)$

